

$^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$ 2011Ga15 $^{38}\text{Si} \rightarrow 2\text{p} + 1\text{n} + ^{35}\text{Mg}$.

2011Ga15: A ^{38}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^9Be target and selected by the A1900 separator at NSCL, MSU. $E(^{38}\text{Si})=83$ MeV/nucleon in the middle of the 376(4)-mg/cm 2 ^9Be secondary target. The reaction leading to ^{35}Mg from ^{38}Si is likely dominated by the two-proton knockout into the continuum of ^{36}Mg and subsequent neutron emission. The reaction residues were selected and identified by the S800 spectrograph using the ΔE -ToF method. The γ rays in coincidence with reaction residues were detected using a 32-fold segmented high-purity germanium detector array (SeGA) at 90° and 37°. Measured Doppler-corrected $E_\gamma(>80$ keV) and deduced levels. Results were compared with Monte Carlo shell-model calculations using the SDPF-M effective interaction that allows for unrestricted mixing of neutron particle-hole configurations across the $N = 20$ gap, and with conventional shell-model calculations with the SDPF-U effective interaction that does not include neutron intruder configurations in the model space.

 ^{35}Mg Levels

$E(\text{level})^\dagger$	J^π	Comments
0	(5/2 $^-$)	J^π : From shell-model calculations with the SDPF-M interaction.
0+x		
0+y		
446+x 5		
621+y 7		
670+y 8		

† From measured E_γ . $x \leq 80$ keV and $y \leq 80$ keV (detection threshold). Shell-model calculations with the SDPF-M interaction predicts a 3/2 $^-$ level at 30 keV.

 $\gamma(^{35}\text{Mg})$

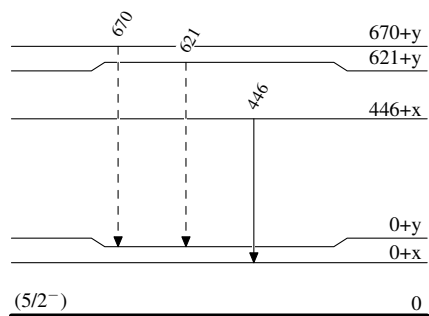
2011Ga15 states that the γ -ray transitions they observed were not in coincidence and depopulate excited states at ≈ 500 and ≈ 700 keV to the ground state or the alleged near-degenerate excited state below 100-keV excitation energy. **2017Mo26** states that the difference of the γ -ray emission between 1n-knockout and 1p1n-removal reaction channels suggests that the 621 and 670-keV γ rays were emitted from two different excited states. Another possible placement of the 621 and 670-keV γ rays: **2011Ga15** states that it might be possible that the two transitions originate from the same level and that their energy difference of 49(11) keV corresponds to the energy spacing of the alleged (5/2 $^-$, 3/2 $^-$) doublet near the ground state. In Adopted Gammas, all γ rays in this dataset are placed to feed the same final levels based on suggestions by **2017Mo26**.

E_γ	$E_i(\text{level})$	E_f	Comments
446 5	446+x	0+x	E_γ : strong γ transition observed in 2011Ga15 ; could correspond to a transition to the g.s. or the shell-model predicted low-lying excited state from any of the shell-model predicted levels at 350 keV with $J^\pi=7/2^-$; 360 keV with $J^\pi=3/2^+$, 400 keV with $J^\pi=1/2^-$, or 560 keV with $J^\pi=7/2^-$.
621 † 7	621+y	0+y	
670 † 8	670+y	0+y	

† Placement of transition in the level scheme is uncertain.

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Legend

Level Scheme-----► γ Decay (Uncertain) $^{35}_{12}\text{Mg}_{23}$